Appendix B. Big-O Notation

In this appendix, we provide a refresher about the RAM model and the renowned big-O mathematical notion, showing you it’s not as bad as you might have heard. Don’t worry, to go through the book we’ll only need a high-level understanding, just the bare minimum to apply it in the algorithm analysis.

B.1 Algorithms and Performance

Describing the performance of an algorithm is not a trivial exercise. You might be used to describe the performance of your code in term of benchmarks, and that might seem straightforward, but if you start thinking about how to describe it effectively, such that your findings can be shared and be meaningful for others, then it raises more issues than you could initially think.

For example, what does performance even mean? It usually implies some kind of measure, so what will it be?

Your first answer might be, you measure the time interval needed to run the algorithm on a certain input. A step further might even be realizing that you might need to average over several inputs, and several runs on the same input, to account for external factors. This would still be a